

Population organization and closed-loop attractors for obstacle avoidance under perturbation

Summary (226 words; 300-word maximum)

Motor systems must adjust ongoing actions when unexpected forces occur, yet continue toward a goal and respect physical constraints. How can a learned controller organize such corrections when obstacles narrow the available route? We asked whether reward-based training alone can produce structured obstacle avoidance and whether the resulting activity offers an organizational comparison with biological motor control. We trained a proximal policy optimization (PPO) controller to move a point mass through a corridor with two obstacles. Its feedforward multilayer perceptron (MLP) actor had two 256-unit hidden layers. We analyzed the activity of a 256-unit hidden-unit population using principal component analysis (PCA), clustering, kinematic tuning, linear decoding, and closed-loop dynamical analysis of the policy coupled to the physics. Moderate lateral perturbations produced path changes under 0.1 m while the controller continued toward the goal. The first two principal components explained 71% of activity variance; activity followed phase-ordered routes shared across perturbation conditions and separated by detour. At the obstacle band, the closed loop had two stable lateral fixed points, one per route ($x^*=3.32$ m and 6.71 m), with similar relaxation times (1.42 s and 1.34 s) and clearances (0.43 m and 0.46 m). These results show that reward learning can yield structured, locally stabilizing routes in this model. The resemblance to biological reaching is organizational, not evidence of neural equivalence; results are from one training seed.

Additional detail: question and approach

Success at reaching a target does not reveal how feedback corrects movement after an unexpected force. The correction must preserve progress toward the goal while respecting the obstacle boundary. We therefore examined route structure and population activity, not only task success. Here, "population" means the 256 artificial units in one hidden layer, not neurons recorded from an animal. The feedforward actor processes each observation independently and has no recurrent memory. Accordingly, the fixed points are not equilibria of recurrent hidden activity (Sussillo & Barak, 2013); they are fixed points of the complete policy-environment closed loop.

Model and perturbation protocol

A proximal policy optimization actor-critic was trained for 3 million environment steps. The actor was a feedforward multilayer perceptron with two 256-unit hidden layers and selected an action from the current state at each timestep. We evaluated a baseline and six perturbation conditions: lateral force impulses in three magnitudes to the left or right, each lasting 14 timesteps. We analyzed activity from a 256-unit hidden layer using principal component analysis, clustering, kinematic tuning, and linear decoding. These analyses characterize activity by movement phase and route and test how much movement information can be read from the population. Motor-cortex recordings motivate this population-level comparison (Churchland et al., 2012).

References

Churchland et al. (2012); Sussillo & Barak (2013); Pruszynski & Scott (2012); Nashed et al. (2014).

Additional detail: closed-loop mechanism and population readout

The policy maps the current state s_t and route cue c to an action; the physics advances the state under perturbation w_t . At $y = 5$ m, we analyze the reduced lateral state $z = (x, v_x)$ separately for each route, with the forward speed at its plateau:

$$s_{t+1} = F_{phys}(s_t, \pi_\theta(s_t; c), w_t) \quad (1)$$

With perturbation absent, a route-specific fixed point satisfies the reduced closed-loop update below; J^* is its local Jacobian. A spectral radius below one indicates local stability for this discrete-time system:

$$z^* = F_{cl}(z^*; c, 0), \rho(J^*) < 1 \quad (2)$$

Both routes had stable, nonoscillatory fixed points at $x^* = 3.32$ m and 6.71 m, within 0.11 m and 0.09 m of completed-episode crossings. The routes cleared the obstacles by 0.43 m and 0.46 m. Relaxation times (1.42 s and 1.34 s) describe model dynamics, not human response latency. Biological reaching also shows graded corrections to unexpected forces while movement continues, supported by coordinated muscle activity (Pruszynski & Scott, 2012; Nashed et al., 2014). This is an organizational comparison, not evidence of neural or muscular equivalence. Results are from one training seed; fixed-point locations vary with checkpoint.

A multinomial logistic-regression decoder classified the perturbation condition from single-timestep hidden activity. Cross-validation held out entire episodes. After the impulse (step 54 onward), mean accuracy was 0.81 (chance = 0.14); before the impulse it was 0.11 , at chance. Errors were mainly between neighboring magnitudes on the same side. The decoder shows that condition information is linearly readable after the perturbation, not that the policy causally uses that representation.

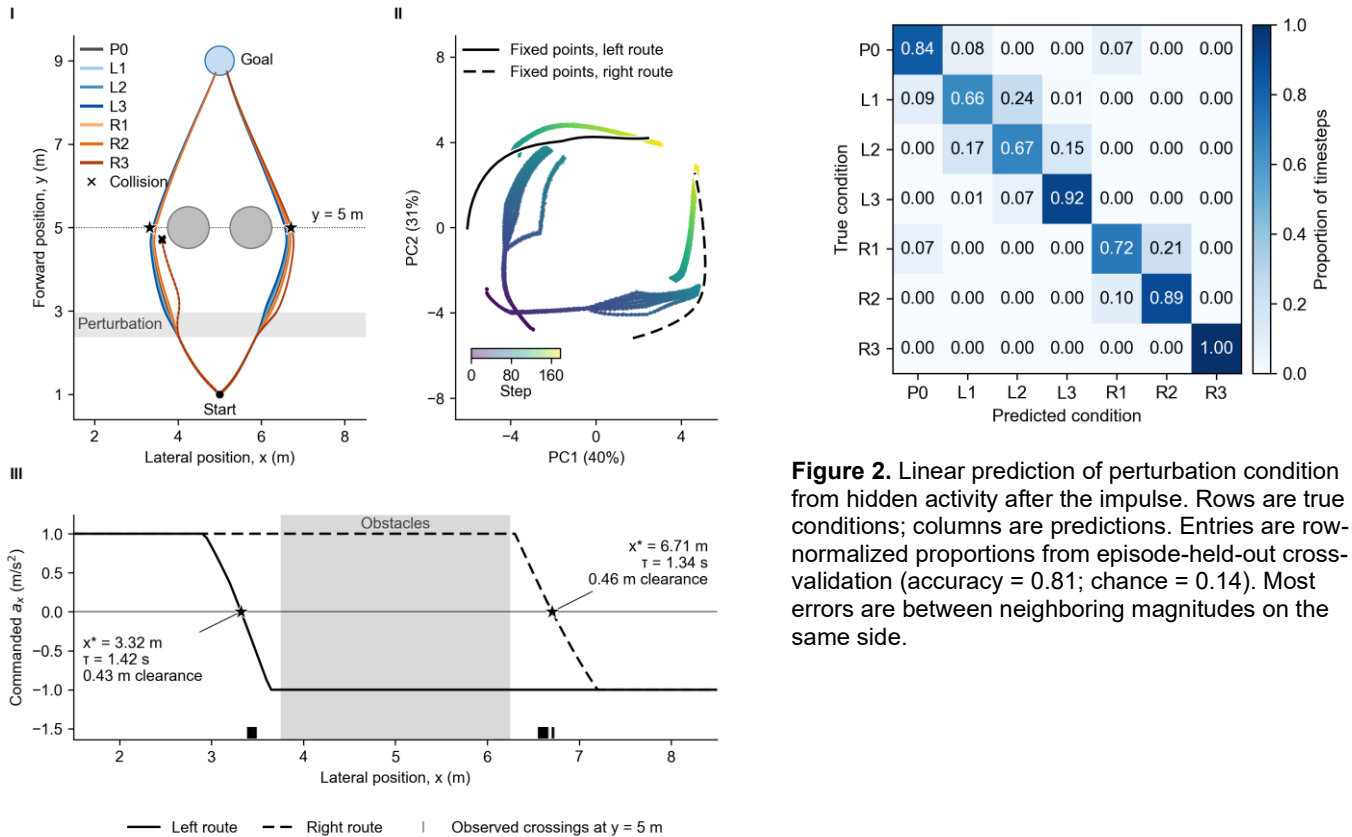


Figure 1. Obstacle avoidance and closed-loop organization. (A) Baseline and perturbation trajectories; the strongest rightward impulse caused collisions on the left route. Stars mark fixed points at $y = 5$ m. (B) Hidden-layer activity projected onto PC1-PC2, colored by step, with fixed-point routes. (C) Commanded lateral acceleration at $y = 5$ m; stable fixed points are negative-slope zero crossings.

Figure 2. Linear prediction of perturbation condition from hidden activity after the impulse. Rows are true conditions; columns are predictions. Entries are row-normalized proportions from episode-held-out cross-validation (accuracy = 0.81 ; chance = 0.14). Most errors are between neighboring magnitudes on the same side.

True condition \ Predicted condition	P0	L1	L2	L3	R1	R2	R3
P0	0.84	0.08	0.00	0.00	0.07	0.00	0.00
L1	0.09	0.66	0.24	0.01	0.00	0.00	0.00
L2	0.00	0.17	0.67	0.15	0.00	0.00	0.00
L3	0.00	0.01	0.07	0.92	0.00	0.00	0.00
R1	0.07	0.00	0.00	0.00	0.72	0.21	0.00
R2	0.00	0.00	0.00	0.00	0.10	0.89	0.00
R3	0.00	0.00	0.00	0.00	0.00	0.00	1.00